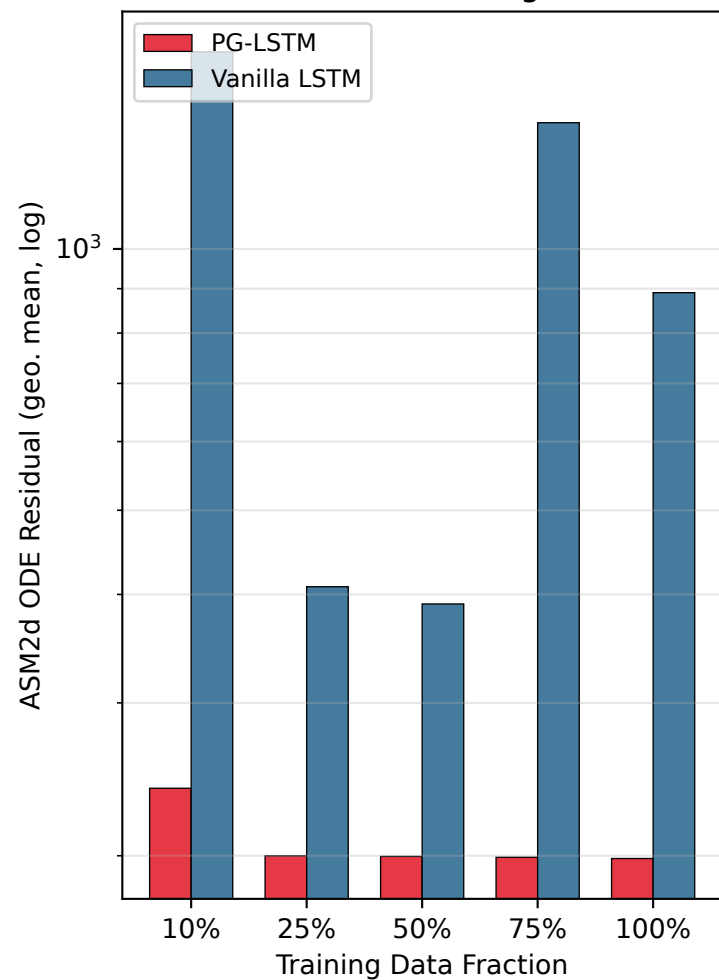
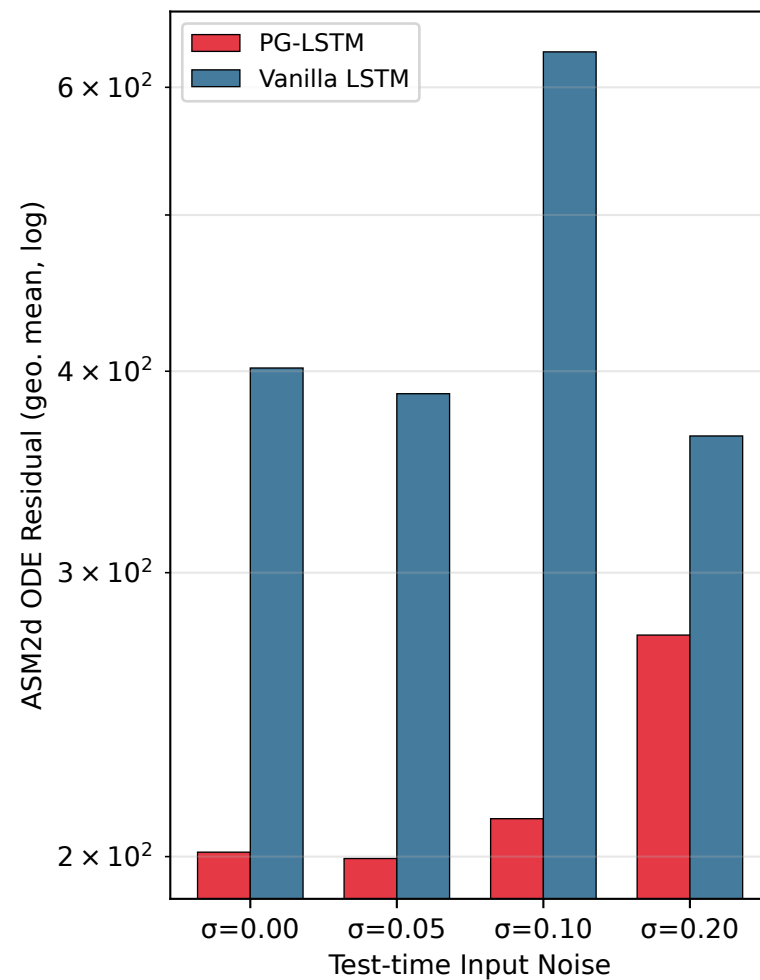


Domain of Utility for PG-LSTM in ASM2d Wastewater Modeling
Equivalent predictive accuracy (TOST $p < 0.05$, see Tab.~\ref{tab:tost}) with consistently lower ASM2d ODE residual across all tested regimes

A — Low-Data Regime



B — Noise Robustness



C — PG-LSTM Mechanistic Consistency Advantage Across Regimes

